Xiao Sha

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EDUCATION

Stony Brook University **Stony Brook, USA**
Ph.D of Electrical and Computer Engineering *Sept. 2017 - PRESENT*

- Research Areas: RF/Analog Circuits, Implantable Neural Devices
- GPA: 3.75/4.00

Xi'an Jiaotong University **Xi'an, China**
Bachelor of Science, Physics & Bachelor of Economics, Finance *Sept. 2013 - Jun. 2017*

- Topic: Synthesis and Characterization of WS_2 and its Heterostructures
- GPA: 3.66/4.00 and 3.46/4.00

Research Projects

Integrated Microsystems Lab **Stony Brook, USA**
Research Assistant *Sept. 2018 - PRESENT*

- **Passive Networks of Tags for Smart Space Based on Backscattering Technology**
  - Designed the passive RF tag in PCB board level that enables 2.5 cm median tracking error with mutil-channel cancellation technology to measure doppler shifts between tags at 915 MHz.
  - Implemented a self-biased common source amplitude shift keying (ASK) demodulator for passive tags in 65 nm technology that features a modulation index as low as 1% with 75 nW power consumption. The design operates at 90 MHz carrier frequency and achieves a data rate of 50-kb/s.
  - A near-zero power fully differential envelop detector first wake-up circuit including 5-stages rectifier, adaptive baseband amplifier, 1 kHz relaxation oscillator and 1.2 V LDO in 180 nm TSMC.
- **Communication Link Design for Distributed mm-sized Free Floating Brain Implants**
  - Designed and optimized a triple-loop inductive link for brain implants. 3D EM simulation was performed to extract four-port impedance parameters initiated with theoretical model. Impedance matching and power transfer efficiency were explored to increase the received signal magnitude.
  - Proposed capacitive coupling link to enable data communication between brain implants with 31 nW at 915 MHz. RC electrical circuit model and two more structures are investigated to reduce leakage current, and the performance is demonstrated better than inductive coupling link.

Graduate Projects

- **Digital Integrated Design:** A pipelined synchronous 8-bit carry select adder works at 2.9 GHz in 45nm technology. Design tradeoffs between power, speed and area were explored. Implementations of schematic and layout were done for Spectre simulation and performance evaluation.
- **Power Grid & Clock Network Analysis:** Sizing and placement of decoupling capacitors to satisfy IR and Ldi/dt noise limitation for power distribution network. A 1 GHz two-level clock distribution network with 20 ps frequency/clock skew and 60 ps clock slew.
- **Computer Architecture Design:** A 12 stages, dual-issue piplined multimedia processor for Sony Cell SPU based on RSIC architectures, including 82 instructions without data hazards (3500 lines).
- **Custom Digital Hardware Generation:** Development of a hardware generation tool for customized matrix-vector multiplication(MVM) systems. The C++ program will generate the corresponding design and an appropriate testbench in System Verilog, based on input data precision, matrix dimension, the desired parallelism, and the amount of pipelining.

- **ATPG Development for VLSI Testing:** Construction of an efficient Automatic Test Pattern Generation (ATPG) based C++ program on PODEM algorithm. Event-driven simulation and smart fault backtracing decision are exploited for acceleration.
- **Mixed-Signal Design:** An 8-bit, 10-MS/s pipelined ADC was designed in 0.6um CMOS. The circuit was built by eight 1.5-bit ADCs for overcoming comparator offset. Performance were optimized and evaluated for integral nonlinearity (INL), effective number of bits (ENOB) and power.
- **Analog Front-end Design:** Design of a front-end charge amplifier and shaper in 45 nm technology for silicon-based particle detectors with maximum 10 kev energy, 4240 dynamic range and 1 mW power consumption.

PUBLICATIONS

- **X. Sha**, Y. Karimi, P. M. Djuri c, M. Stanacevic, and S. R. Das,"Study of Mm-Sized coil to coil backscatter based communication link," 2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York City, NY, USA, 2019, pp. 1124-1129.
- **X. Sha**, Y. F. Huang, T. T. Wan, Y. Karimi, P. M. Djuri c, M. Stanacevic, and S. R. Das,"A Self-Biased Low Modulation Index ASK Demodulator for Implantable Devices." ISCAS 2020
- Ahmad, A., Y. F. Huang, **X. Sha**, A. Athalye, M. Stanacevic, and S. R. Das and P. M. Djuri 'c " On Measuring Doppler shifts between tags in a Backscattering Tag-to-Tag Network with applications in tracking" ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2020, pp. 9055-9059.
- **X. Sha**, P. Y. Zheng, Y. Karimi, and M. Stanacevic "Capacitive Link for Data Communication Between Free Floating mm-sized Brain Implants" 2021 IEEE International Symposium on Medical Measurements and Applications (MeMeA) (Submitted)
- Zhang, L., **X. Sha**, Q. K. Fan, L. Han, Y. D. Yin and C. B. Gao (2017). "Gold nanoshurikens with uniform sharp tips for chemical sensing by the localized surface plasmon resonance." Nanoscale 9(43): 17037-17043.
- Zhang, L., Q. K. Fan, **X. Sha**, P. Zhong, J. Zhang, Y. D. Yin and C. B. Gao (2017)."Self-assembly of noble metal nanoparticles into sub-100 nm colloidosomes with collective optical and catalytic properties." Chemical Science 8(9): 6103-6110.
- Li, C. Y., M. S. Ding, Y. Yang, P. C. Zhang, Y. Li, Y. C. Wang, L. C. Huang, P. J. Yang, M. Wang, **X. Sha**, Y. M. Xu, C. W. Guo, Z. W. Shan(2016)"Portrait and Classification of Individual Haze Particulates." Journal of Environmental Protection 7(10): 1355

Industrial Experience

Skysys Intelligent Technology **Shanghai, China**
Embedded System Engineer Intern *Jun. 2018 - Aug. 2018*

- Developed a step motors driver program with uniform speed increasing to protect the initial phase for drone hangar gate based on STM32F407.
- Improved control and operation system, offered data transfer between MCU and Linux mother board, optimized wire arrangement and power supply.
- Prepared and wrote two patents about the drone landing platform and battery auto-charging block.

Professional activities

- **Student Member:** IEEE, Circuits and Systems Society
- **Reviewer:**
  - IEEE Biomedical Circuits and Systems Conference (BIOCAS 2019)
  - Engineering in Medicine and Biology Conference (EMBC 2019, 2020)
  - International Symposium on Circuits & Systems (ISCAS 2020,2021)

Skills

Tools: Cadence Virtuoso, ModelSim, HSpice, Spectre, Quartus, Altium Designer, Ansys HFSS, Advanced Design System (ADS), Origin

Language: System Verilog, Python, Matlab, C/C++, HTML, Markdown